Tableau Basics

Q1. What is Tableau, and why is it preferred over spreadsheets?

Tableau is a data visualization and business intelligence tool that lets users connect to data sources, and build interactive charts, dashboards, and reports without writing complex code. It uses a drag-and-drop interface: fields are placed onto shelves (Rows, Columns, Filters, Marks) and Tableau automatically renders an appropriate visualization.

Why Tableau is preferred over traditional spreadsheet tools:

- **Interactivity:** Dashboards allow filtering, drilling down, and hovering for details, unlike static spreadsheet charts.
- **Speed of analysis:** Drag-and-drop fields build charts instantly instead of writing formulas or pivot tables manually.
- **Handles larger and varied data:** Connects directly to databases, cloud sources, and big data without manual copy-paste.
- **Better visual storytelling:** Offers maps, heat maps, and advanced chart types that spreadsheets cannot easily produce.
- **Real-time / live connections:** Can refresh from live data sources so dashboards stay current automatically.

Industries where Tableau is commonly used:

- **Retail:** Analyzing sales, profit, and inventory across regions and product categories to identify top-performing stores and products.
- **Healthcare:** Tracking patient outcomes, hospital resource utilization, and disease trends to support clinical and operational decisions.

Example use case (Retail): A retail chain uses Tableau to build a sales dashboard showing revenue and profit by region and category. Store managers use it to instantly spot underperforming regions and adjust promotions or inventory allocation, something that would take much longer using manual spreadsheet reports.

Q2. Fields in the Sample – Superstore dataset

a) Main types of fields available:

- **Time-based (Date) fields:** e.g., Order Date, Ship Date
- **Geographical fields:** e.g., Country, State, City, Region, Postal Code
- **Numerical (quantitative) fields:** e.g., Sales, Profit, Quantity, Discount

- **Categorical (qualitative) fields:** e.g., Category, Sub-Category, Segment, Ship Mode

b) Four Dimensions and Four Measures:

Category	Sales
Sub-Category	Profit

Region	Quantity
Customer Name	Discount

Dimensions are qualitative fields shown in blue in the Data Pane, used to categorize or slice data. Measures are quantitative, numeric fields shown in green, used to aggregate and quantify data.

Q3. Bar Chart – Total Sales by Category

- **Rows shelf:** Category (dimension) is placed on Rows — this creates one row per category (Furniture, Office Supplies, Technology).
- **Columns shelf:** Sales (measure) is placed on Columns — this determines the length of each bar.

Aggregation applied: Tableau applies SUM by default to numeric measures, so the chart shows SUM(Sales) for each category.

Steps: Drag **Category** to the Rows shelf and **Sales** to the Columns shelf (or vice versa for a vertical bar chart). Tableau automatically generates a bar chart because one field is a dimension and the other is a continuous measure. The Marks card will show 'Automatic', which defaults to a Bar mark type in this scenario.

Q4. Comparing Sales and Profit; Profit by Sub-Category

Comparing Sales and Profit together reveals whether high sales actually translate into high profitability. A sub-category can have strong sales but low or negative profit due to heavy discounting or high costs — this comparison helps identify products that are 'selling well but not making money,' which is critical for pricing, discounting, and inventory decisions.

Chart to analyze Profit by Sub-Category:

- Place **Sub-Category** (Dimension) on the Rows shelf.

- Place **Profit** (Measure) on the Columns shelf.
- Sort the bars descending to quickly identify the most and least profitable sub-categories.
- Optionally add a reference line at zero, and color the bars by a calculated field (Profit >= 0) to instantly highlight loss-making sub-categories in red.

Q5. Line Chart – Sales Trend over Order Date

- Drag **Order Date** to the Columns shelf and **Sales** to the Rows shelf. Tableau automatically builds a line chart for a continuous date field vs. a measure.

Date level chosen: Month (or Month/Year via Order Date's date hierarchy). Month-level granularity is usually the best balance: Year is too coarse to reveal seasonal patterns, while Day/Week can be too noisy. Right-clicking the Order Date pill and selecting the 'Month' level (continuous) shows month-over-month trends while still highlighting seasonality (e.g., holiday sales spikes).

Why line charts suit time-based analysis:

- They clearly show the direction and rate of change over time (increasing, decreasing, or seasonal patterns).
- The continuous connecting line makes trends, peaks, and dips easy to spot at a glance.
- They allow easy comparison of multiple trends (e.g., by adding Category to color) on the same time axis.

Q6. Basic Filter — Region = West, Category = Technology

Shelf used: Filters shelf.

Steps:

- Drag the **Region** field from the Data Pane onto the **Filters** shelf.
- In the filter dialog, select only **"West"** from the list of regions and click OK.
- Drag the **Category** field onto the **Filters** shelf as well.
- In its filter dialog, select only **"Technology"** and click OK.
- The view now updates to display only records where Region = West AND Category = Technology, since multiple filters on the Filters shelf combine with an AND condition.

Q7. Geographical Analysis and Map – Sales by State

Geographical analysis is important because business performance often varies significantly by

location. Mapping data lets decision-makers instantly identify high-performing and underperforming regions, plan logistics and warehouse placement, target marketing by geography, and allocate resources where they will have the greatest impact — patterns that are far harder to notice in a plain table of numbers.

Steps to create the map:

- Double-click the **State** field (or drag it onto the view). Since State is recognized as a geographic role, Tableau automatically generates a map with Latitude/Longitude and plots each state.
- Drag **Sales** onto the Color mark (or Size mark) on the Marks card.
- Tableau shades (or sizes) each state according to its SUM(Sales), producing a filled or symbol map showing sales distribution across the U.S.
- Optionally, change the map type to 'Filled Map' under the Marks card dropdown for a heat-map style view by state.

Q8. Worksheet vs. Dashboard

	art built from one data source (e.g., one). A combination of multiple worksheets, d filters arrange
	d refine an individual visualization. Used to ete, interactive story combining several
Limited to one chart at a time.	Allows cross-filtering — clicking on on

Why dashboards are more useful for decision-makers: Dashboards consolidate multiple related metrics (sales, profit, region, trend) into a single interactive screen. Executives can see the full picture at a glance, apply a filter once and have it update every chart, and drill into details without switching between separate worksheets — making decision-making faster and more holistic.

Q9. Tableau Prep and Data Cleaning Steps

Tableau Prep is a separate data preparation tool from Tableau (Tableau Prep Builder) used to clean, shape, combine, and reshape data visually — using a flow of drag-and-drop steps — before it is loaded into Tableau Desktop for analysis.

Why it's important in real-world data projects:

- Real-world data is rarely clean — it often has missing values, duplicates, inconsistent formats, or wrong data types.

- Tableau Prep lets analysts fix these issues visually (without writing SQL/code) and see the impact of each step immediately.
- It creates a repeatable 'flow' that can be re-run automatically whenever new data arrives, saving time on recurring reports.

How to perform the following (in steps, conceptually):

Remove null values:

- In the Prep flow, click on the field/column showing null values in the profile pane.
- Use the 'Filter' option and choose to exclude Null values, or right-click the null value in the data grid and select 'Exclude'.
- Add a Clean step that filters out rows where the key field is null before it moves further down the flow.

Remove duplicate records:

- Add a 'Clean Step' to the flow.
- Select the relevant fields (e.g., Order ID, Customer, Product) and use the Group & Replace or a De-duplicate step to identify identical rows.
- Remove/aggregate the redundant rows so each record is unique.

Change incorrect data types:

- Click the data type icon (e.g., '#', 'Abc', calendar icon) at the top of the column header.
- Select the correct type from the dropdown — for example, change a Sales column stored as text ('Abc') to a Number (decimal), or a date stored as text to a proper Date type.
- Tableau Prep will attempt an automatic conversion and flag any values it could not convert, which can then be manually corrected or excluded.

Q10. Retail Manager Dashboard Scenario

a) Charts to use:

- **Which category generates the highest profit?** — Bar chart of Profit by Category (Category on Rows/Columns, Profit as the measure), sorted descending.
- **Which region performs poorly?** — Bar chart or Map of Profit/Sales by Region, sorted ascending to surface the weakest region, optionally colored red for negative profit.
- **How does sales change over time?** — Line chart of SUM(Sales) over Order Date (Month level) to show the trend.

b) Key insight management could derive:

For example, the Technology category may generate the highest profit while the Furniture category, despite decent sales, often shows thin or negative profit margins due to heavy discounting. Similarly, one region (e.g., Central) may consistently underperform on profit compared to others, and sales typically show a seasonal spike toward the end of the year (Nov–Dec).

c) How this helps business decision-making:

- Identifying the most profitable category helps management focus marketing spend and inventory investment where returns are highest.
- Spotting an underperforming region enables targeted interventions — revised pricing, better staffing, or focused promotions in that area.
- Understanding seasonal sales trends helps with inventory planning, staffing levels, and cash-flow forecasting ahead of peak periods.
- Overall, combining these three views on one dashboard lets a manager make faster, data-driven decisions instead of relying on gut feeling.